

OOPS, Design Patterns, SOLID Principles and SQL Interview Questions (50 Questions)

Topics Covered:

- OOP Concepts
- SOLID Principles
- Design Patterns
- SQL Basics
- Joins
- Group By
- Aggregate Functions
- Subqueries
- SQL Interview Problems

Suitable for 0–3 years experience.

OOPS, SOLID and Design Pattern Questions (25)

1. What is Object-Oriented Programming (OOP)?

Answer

Object-Oriented Programming is a programming paradigm based on objects and classes. It helps organize code using concepts like encapsulation, inheritance, polymorphism, and abstraction. OOP improves reusability, maintainability, and scalability.

2. What is a class and object?

Answer

A class is a blueprint used to create objects, while an object is an instance of a class. Classes define properties and methods, and objects represent real-world entities.

3. What is encapsulation?

Answer

Encapsulation is the process of binding data and methods together inside a class while restricting direct access to internal data. It improves data security and maintainability.

4. What is inheritance?

Answer

Inheritance allows one class to acquire properties and methods from another class. It improves code reuse and supports hierarchical relationships.

5. What is polymorphism?

Answer

Polymorphism allows methods or objects to behave differently based on context. Method overloading and method overriding are common examples.

6. What is abstraction?

Answer

Abstraction hides implementation details and exposes only essential functionality. It simplifies complex systems and improves maintainability.

7. Difference between abstract class and interface?

Answer

An abstract class can contain both abstract and concrete methods, while an interface mainly defines behavior contracts. Java interfaces also support multiple inheritance.

8. Difference between method overloading and overriding?

Answer

Method overloading occurs within the same class using different parameters, while overriding occurs when a child class redefines a parent class method.

9. What is constructor?

Answer

A constructor is a special method used to initialize objects. It has the same name as the class and does not have a return type.

10. What is this keyword?

Answer

The this keyword refers to the current object of the class. It is commonly used to differentiate instance variables from local variables.

11. What is super keyword?

Answer

The super keyword refers to the parent class object. It is used to access parent class methods, constructors, and variables.

12. What is composition?

Answer

Composition represents a HAS-A relationship between classes. It improves flexibility and loose coupling compared to inheritance.

13. What is aggregation?

Answer

Aggregation is a weaker form of composition where objects can exist independently of each other. It also represents a HAS-A relationship.

14. What are SOLID principles?

Answer

SOLID is a set of five object-oriented design principles used to create maintainable and scalable applications. These principles reduce tight coupling and improve flexibility.

15. What is Single Responsibility Principle?

Answer

Single Responsibility Principle states that a class should have only one reason to change. Each class should focus on a single responsibility.

16. What is Open Closed Principle?

Answer

Open Closed Principle states that software entities should be open for extension but closed for modification. Existing code should not be changed unnecessarily.

17. What is Liskov Substitution Principle?

Answer

Liskov Substitution Principle states that child classes should be replaceable with parent classes without affecting correctness.

18. What is Interface Segregation Principle?

Answer

Interface Segregation Principle states that clients should not be forced to implement unnecessary methods. Smaller interfaces are preferred over large interfaces.

19. What is Dependency Inversion Principle?

Answer

Dependency Inversion Principle states that high-level modules should depend on abstractions rather than concrete implementations.

20. What is Singleton Design Pattern?

Answer

Singleton pattern ensures only one object instance exists throughout the application. It is commonly used for configuration and logging.

21. What is Factory Design Pattern?

Answer

Factory pattern creates objects without exposing object creation logic to the client. It improves flexibility and loose coupling.

22. What is Builder Design Pattern?

Answer

Builder pattern is used to create complex objects step by step. It improves readability and avoids large constructors.

23. What is Strategy Design Pattern?

Answer

Strategy pattern allows selecting algorithms dynamically at runtime. It promotes flexibility and avoids large conditional statements.

24. What is Observer Design Pattern?

Answer

Observer pattern defines one-to-many dependency between objects. When one object changes state, dependent objects are notified automatically.

25. Difference between tight coupling and loose coupling?

Answer

Tight coupling occurs when classes depend heavily on each other, making changes difficult. Loose coupling reduces dependencies and improves flexibility and maintainability.

SQL Questions (25)

26. What is SQL?

Answer

SQL stands for Structured Query Language and is used to interact with relational databases. It is used for storing, retrieving, updating, and deleting data.

27. Difference between DELETE, TRUNCATE, and DROP?

Answer

DELETE removes rows and can be rolled back. TRUNCATE removes all rows quickly without rollback in many databases. DROP removes the entire table structure.

28. What is primary key?

Answer

A primary key uniquely identifies each record in a table. It cannot contain null or duplicate values.

29. What is foreign key?

Answer

A foreign key creates a relationship between two tables. It references the primary key of another table.

30. Difference between WHERE and HAVING?

Answer

WHERE filters rows before grouping, while HAVING filters grouped data after GROUP BY. WHERE cannot use aggregate functions directly, but HAVING can.

Example:

```
SELECT dept_id, COUNT(*)  
FROM employee  
WHERE salary > 30000  
GROUP BY dept_id  
HAVING COUNT(*) > 2;
```

31. What is GROUP BY?

Answer

GROUP BY groups rows with similar values together and is commonly used with aggregate functions such as COUNT, SUM, AVG, MIN, and MAX.

Example:

```
SELECT dept_id, COUNT(*)  
FROM employee  
GROUP BY dept_id;
```

Answer

GROUP BY groups rows with similar values together. It is commonly used with aggregate functions like COUNT and SUM.

32. What are aggregate functions?

Answer

Aggregate functions perform calculations on multiple rows and return a single result. Common aggregate functions include COUNT, SUM, AVG, MIN, and MAX.

Example:

```
SELECT COUNT(*), AVG(salary), MAX(salary)  
FROM employee;
```

33. Difference between INNER JOIN and LEFT JOIN?

Answer

INNER JOIN returns only matching rows from both tables, while LEFT JOIN returns all rows from the left table and matching rows from the right table.

Example:

```
SELECT e.name, d.dept_name  
FROM employee e  
INNER JOIN department d  
ON e.dept_id = d.id;
```



```
SELECT e.name, d.dept_name
FROM employee e
LEFT JOIN department d
ON e.dept_id = d.id;
```

Answer

INNER JOIN returns matching rows from both tables. LEFT JOIN returns all rows from the left table and matching rows from the right table.

34. What is normalization?

Answer

Normalization is the process of organizing database tables to reduce redundancy and improve consistency.

35. What is denormalization?

Answer

Denormalization combines tables to improve read performance by reducing joins.

36. What is index in SQL?

Answer

An index improves query performance by allowing faster data retrieval from tables. However, indexes increase storage usage and slightly slow down insert and update operations.

Example:

```
CREATE INDEX idx_employee_name
ON employee(name);
```

37. Difference between clustered and non-clustered index?

Answer

Clustered index determines the physical order of data in the table, while non-clustered index creates a separate structure pointing to table rows. A table can have only one clustered index but multiple non-clustered indexes.

Example:

```
CREATE CLUSTERED INDEX idx_emp_id  
ON employee(id);
```

Answer

Clustered index determines physical storage order of data, while non-clustered index stores separate index structures.

38. What is subquery?

Answer

A subquery is a query written inside another SQL query. It is commonly used for filtering, comparisons, and retrieving intermediate results.

Example:

```
SELECT name  
FROM employee  
WHERE salary > (  
    SELECT AVG(salary)  
    FROM employee  
);
```

39. What is correlated subquery?

Answer

A correlated subquery depends on values from the outer query and executes repeatedly for each row processed by the outer query.

Example:

```
SELECT e1.name, e1.salary
FROM employee e1
WHERE salary > (
    SELECT AVG(e2.salary)
    FROM employee e2
    WHERE e1.dept_id = e2.dept_id
);
```

Answer

A correlated subquery depends on values from the outer query and executes repeatedly for each row.

40. How to find duplicate records?

Answer

Duplicate records can be found using GROUP BY and HAVING COUNT(*) > 1.

Example:

```
SELECT name, COUNT(*)
FROM employee
GROUP BY name
HAVING COUNT(*) > 1;
```

41. How to find second highest salary?

Answer

Second highest salary can be found using ORDER BY with LIMIT/OFFSET or subqueries.

Example:

```
SELECT MAX(salary)
FROM employee
WHERE salary < (
    SELECT MAX(salary) FROM employee
);
```

42. How to find third highest salary?

Answer

Third highest salary can be found using nested subqueries or LIMIT/OFFSET.

Example:

```
SELECT DISTINCT salary
FROM employee
ORDER BY salary DESC
LIMIT 1 OFFSET 2;
```

43. How to find employees without department?

Answer

LEFT JOIN with NULL checks can identify employees without matching department records.

Example:

```
SELECT e.*
FROM employee e
LEFT JOIN department d
ON e.dept_id = d.id
WHERE d.id IS NULL;
```

44. What is UNION and UNION ALL?

Answer

UNION combines results from multiple queries and removes duplicate rows, while UNION ALL combines results without removing duplicates. UNION ALL is generally faster because it skips duplicate checking.

Example:

```
SELECT name FROM employee
UNION
SELECT name FROM manager;
```

45. What is difference between CHAR and VARCHAR?

Answer

CHAR stores fixed-length strings, while VARCHAR stores variable-length strings. VARCHAR is more memory efficient for varying text sizes.

Example:

```
name CHAR(10)
city VARCHAR(100)
```

Answer

CHAR stores fixed-length strings, while VARCHAR stores variable-length strings. VARCHAR is more memory efficient.

46. What is view in SQL?

Answer

A view is a virtual table created using a SQL query. Views simplify complex queries and improve security by restricting direct table access.

Example:

```
CREATE VIEW high_salary_emp AS
SELECT *
FROM employee
WHERE salary > 50000;
```

47. What is stored procedure?

Answer

A stored procedure is a precompiled collection of SQL statements stored inside the database. It improves performance, reusability, and centralized business logic.

Example:

```
CREATE PROCEDURE getEmployees()
BEGIN
```

```
SELECT * FROM employee;  
END;
```

Answer

A stored procedure is a precompiled collection of SQL statements stored in the database. It improves reusability and performance.

48. What is ACID property?

Answer

ACID properties ensure reliable database transactions. They stand for Atomicity, Consistency, Isolation, and Durability.

49. What is transaction?

Answer

A transaction is a group of SQL operations executed as a single unit of work. Transactions ensure data consistency and reliability using ACID properties.

Example:

```
START TRANSACTION;  
UPDATE account SET balance = balance - 100 WHERE id = 1;  
UPDATE account SET balance = balance + 100 WHERE id = 2;
```

50. What is commit and rollback?

Answer

COMMIT permanently saves transaction changes to the database, while ROLLBACK undoes uncommitted changes if an error occurs.

Example:

```
COMMIT;  
ROLLBACK;
```

Answer

COMMIT permanently saves transaction changes, while ROLLBACK undoes uncommitted changes.